Workshop 2, Nyi Nyi Naing, 22 May 2026

Automated Data Analysis and Reporting

Relationship Between GDP per Capita and Life Expectancy (2020)

Report based on the uploaded country-level datasets.

1. Data check and 2020 subset

GDP per capita: 193 country rows; life expectancy: 194 country rows. Both datasets are in wide format with one row per country and one column per year. There are no duplicate country codes in either file.

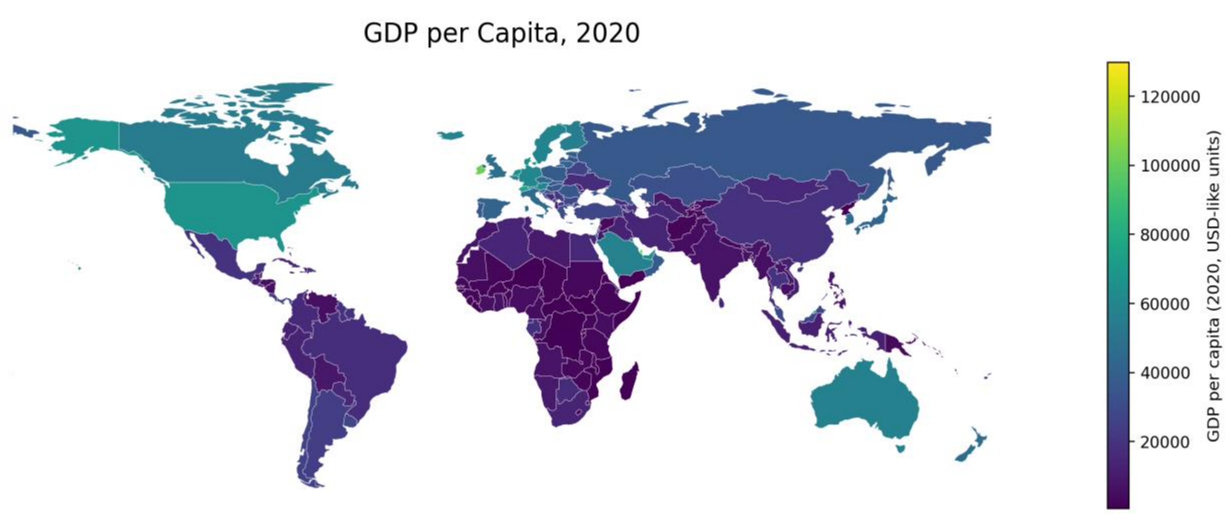
For 2020, GDP per capita has 193 values and life expectancy has 194 values. The merged 2020 scatter data includes 193 countries.

Country	GDP per capita 2020	Life expectancy 2020
Afghanistan	2769.69	64.6
Angola	7556.97	62.8
Albania	14662.80	77.4
Andorra	55488.49	82.7
UAE	65784.68	79.5

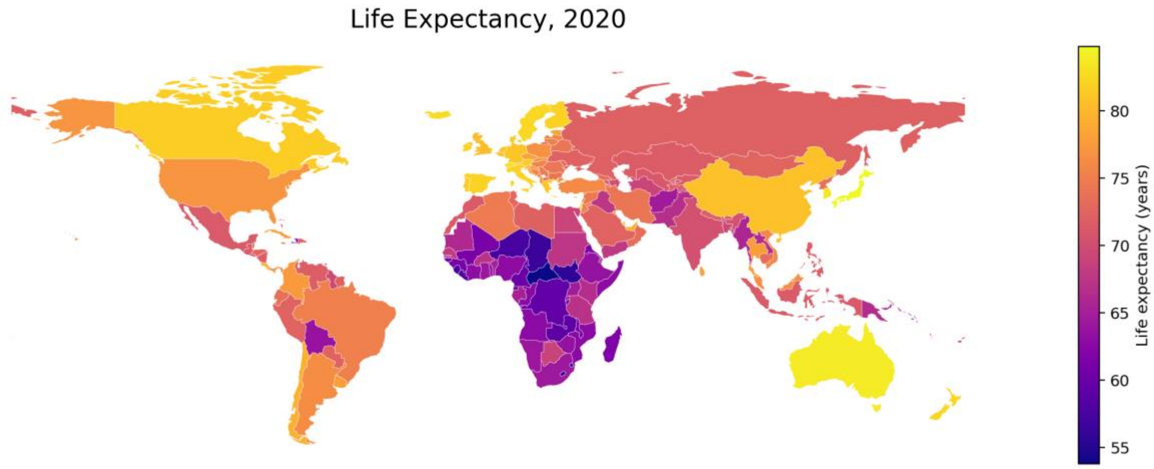
The base world geometry does not include a few territories, so the maps omit Falkland Islands, Greenland, New Caledonia, Puerto Rico, Taiwan, and Western Sahara.

2. World maps for 2020

GDP per Capita by country



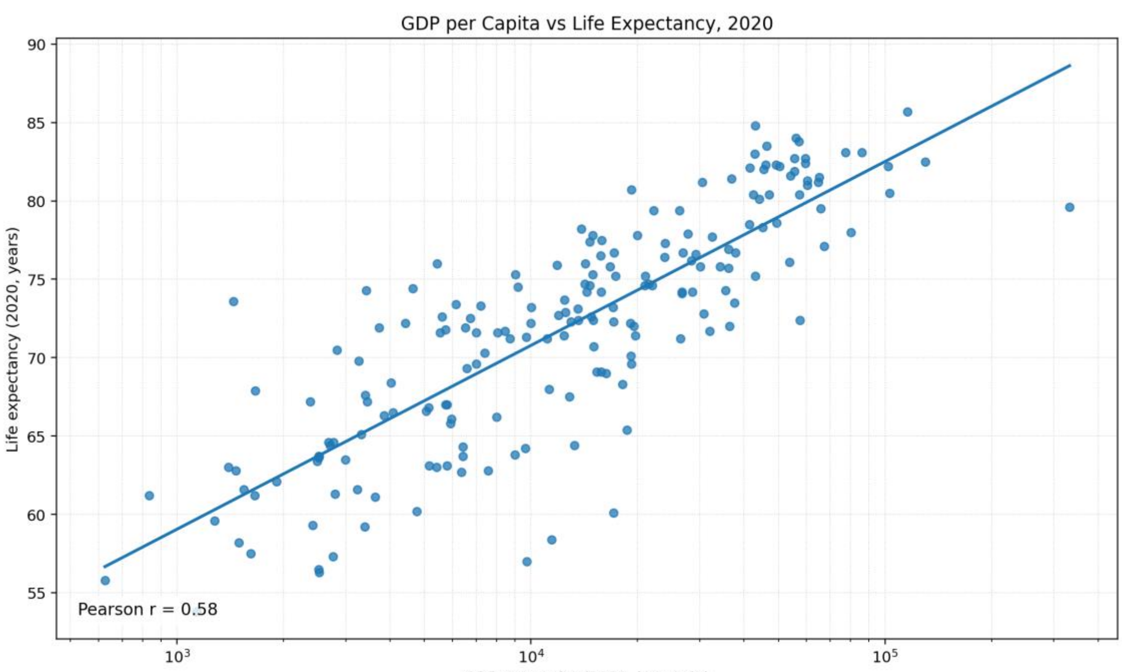
Life Expectancy by country



3. Scatterplot

GDP per Capita Vs Life expectancy

A log scale on GDP per capita makes the country pattern easier to see across the full income range.



4. Interpretation

The relationship is positive: higher GDP per capita is associated with higher life expectancy. The 2020 Pearson correlation is 0.58, so the association is moderate to strong. The scatterplot is not perfectly linear, and life expectancy gains flatten somewhat at higher income levels; this suggests that income matters, but health systems, public policy, education, and demographics also play major roles.

5. Brief Reflection on Using AI Tools

The AI made it easy to organize the data, create maps and graphs, and identify the relationship between GDP per capita and life expectancy quickly.

The main challenge was making sure the datasets were clean and correctly formatted before analysis.

It was surprising how fast the AI could generate visualizations and provide interpretations from raw data.

I learned that AI is very useful for data analysis and visualization, but human checking and interpretation are still important for accurate results.